INFORMATION TO KEEP WITH YOU

My complete treatment handbook

Nivolumab, pemetrexed and carboplatin before surgery

Understand your treatment. Know what to expect. Keep your plan to hand.

Displayed calendar
21 days
See the active treatment phase

Treatment stages
2
Explained in the treatment reference

Regimen source
NCCP 00849
Version 2

Up to three 21-day cycles. Day 1: nivolumab, pemetrexed and carboplatin. Then surgical multidisciplinary team (MDT) reassessment. The five listed backbones are alternatives; this guide covers only your selected combination.

Treatment stages

01 **Before surgery**
Up to three cycles

02 **Surgical reassessment**
After treatment

Your team confirms the active stage and dates. The full treatment guide explains each step.

Inside your handbook

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My name

Hospital / oncology team

Use the urgent instructions and contact numbers in your treatment guide. Your team must replace example numbers before patient use. Your individual treatment plan takes precedence.

CHAPTER 01

Understanding your treatment

IN THIS CHAPTER

Why this treatment is offered, what its medicines do, and how the agreed plan fits your care.

When to get help

Emergency: call 112 / 999

Severe breathing difficulty, collapse, seizure or severe chest pain: call emergency services. Do not wait for an online answer.

Contact your oncology service immediately

Temperature 37.5°C or higher, 36°C or lower, shivering or suddenly feeling unwell — even with a normal temperature. Contact your oncology team urgently for fever, feeling suddenly unwell, new breathlessness, chest pain, persistent diarrhoea, severe weakness, confusion, reduced urine or a severe rash. Follow your local emergency instructions; do not wait for the next appointment.

Follow the urgent contact instructions from your treating hospital. If you cannot reach the team, seek urgent medical assessment. Outside clinic hours, use your instructed out-of-hours pathway or the emergency department.

This page does not diagnose symptoms or decide whether it is safe to stay at home. You do not need to know which medicine caused a symptom before calling.

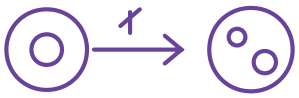
Three medicines. Different jobs.

This treatment is given before planned surgery for resectable non-small-cell lung cancer at high risk of returning. It aims to treat the cancer before the operation and reduce the chance of recurrence. Surgery and benefit are not guaranteed.

IMMUNOTHERAPY

Nivolumab

Opdivo



Blocks the PD-1 immune checkpoint

Nivolumab binds PD-1 on T cells. It prevents signals that can reduce the immune attack on cancer. This can also let immune cells inflame healthy tissues. It is infused every 21 days in this plan.

T cell → PD-1 blocked → Immune response

What to watch for

New breathing, bowel, skin, hormone or other symptoms need prompt advice during or after treatment.

ANTIFOLATE CHEMOTHERAPY

Pemetrexed

Alimta



Interrupts deoxyribonucleic acid (the cell's genetic material) (DNA) building-block production

Pemetrexed blocks several enzymes that use folate to make the building blocks of DNA and RNA. This can slow or stop cancer-cell growth. Healthy cells can also be affected. Prescribed folic acid and vitamin B12 reduce important toxicities and are part of treatment.

Folate-dependent enzymes blocked → DNA building blocks reduced → Cell growth disrupted

What to watch for

Take the prescribed folic acid and attend for B12. Report mouth sores, diarrhoea, a rash, breathlessness or reduced urine; check before taking ibuprofen or other non-prescription medicines.

CHEMOTHERAPY

Carboplatin

Platinum chemotherapy



Damages DNA

Carboplatin forms links in DNA that hinder copying and repair. Healthy tissues can also be harmed. Its dose depends on target exposure (AUC) and measured or estimated kidney function; AUC is not a fixed number of milligrams. There are up to three planned doses, one in each neoadjuvant cycle before surgical review.

DNA → Platinum links → Copying disrupted

What to watch for

Report fever, bleeding, nausea, hearing changes or numbness. Tell the nurse immediately about possible allergy during the infusion.

The drawings are simplified explanations of drug action, not anatomical pictures or a prediction of response.

Your treatment before surgery

Why these medicines?

Nivolumab helps immune cells recognise cancer; chemotherapy damages dividing cells. Your prescribed backbone is pemetrexed and carboplatin.

Evidence and uncertainty

In a randomized study, adding nivolumab to chemotherapy before surgery improved survival compared with chemotherapy alone. Your team considers the exact cancer, test results, fitness and preferences.

Alternatives

Discuss surgery and other suitable treatments, clinical trials and the option of not proceeding; choices depend on the multidisciplinary assessment.

Why treatment is offered before surgery

The trial population was broader than NCCP eligibility. Trial averages cannot predict your own outcome or compare these five chemotherapy options.

CHAPTER 02

Your schedule, tests and appointments

IN THIS CHAPTER

Your treatment stages, checks and practical preparations, from before treatment through follow-up.

Induction → review → maintenance

Treatment stages

01 **Before surgery**
Up to three cycles

02 **Surgical reassessment**
After treatment

Your team confirms the active stage and dates. The full treatment guide explains each step.

This shows treatment timing, not guaranteed symptom-free days. Your team confirms the active phase and dates.

UP TO THREE CYCLES
Before surgery

1 2 3

Up to three 21-day cycles. Day 1: nivolumab, pemetrexed and carboplatin. Then surgical multidisciplinary team (MDT) reassessment. The five listed backbones are alternatives; this guide covers only your selected combination.

AFTER TREATMENT
Surgical reassessment

MDT review

The surgical team reviews scans, fitness, recovery and timing. This regimen does not automatically continue nivolumab after surgery.

Maintenance means ongoing treatment, not treatment without side effects. Your plan can change after discussion.

BASELINE → TREATMENT → FOLLOW-UP

Tests and decisions across the treatment course

These are relative stages, not equal time intervals. Additional tests depend on symptoms and the selected chemotherapy source.

The team arranges and reviews tests. This guide shows expected timing; it does not record completed appointments. NCCP 00849 v2 · official regimen · NCCP 00318 v6 · selected chemotherapy

THE WHOLE TREATMENT JOURNEY

Nivolumab, pemetrexed and carboplatin before surgery: treatment and checks

Neoadjuvant non-small-cell lung cancer (NSCLC) · selected pemetrexed / carboplatin backbone

Up to three 21-day cycles. Day 1: nivolumab, pemetrexed and carboplatin. Then surgical MDT reassessment. The five listed backbones are alternatives; this guide covers only your selected combination.

Read from left to right. Columns show stages, not equal lengths of time. This is a general plan: no dates or personal information are needed. Your team confirms which steps apply and arranges appointments.

Treatment, tests and reviews across the course

Care / check	Preparation	Cycle 1	Cycle 2	Cycle 3	Surgical review	Follow-up
Treatment NCCP 00849 · COMPANION	Confirm selected backbone	Day 1 all three	Repeat if appropriate	Final planned cycle	No automatic systemic anticancer therapy (SACT) continuation	Separate plan if further treatment needed
Blood and organ tests NCCP 00849 · COMPANION	full blood count (FBC), kidney, liver, glucose, thyroid and viral screen	FBC, kidney, liver, glucose	Repeat before treatment	Repeat before treatment	Recovery and fitness checks	Follow unresolved abnormalities
Immune monitoring NCCP 00849 · COMPANION	Document baseline symptoms	Symptoms before every dose	Thyroid every 3–6 weeks	Cortisol if clinically indicated	Report delayed symptoms	Vigilance at least five months

Care / check	Preparation	Cycle 1	Cycle 2	Cycle 3	Surgical review	Follow-up
Support and decisions NCCP 00849 · COMPANION	Take the prescribed folic acid, vitamin B12 and rash-prevention steroid plan; continue folic acid for 21 days after the final pemetrexed dose. Carboplatin dosing uses a verified kidney-function measurement; report dehydration or a change in health.	Confirm written prevention plan	Review tolerability	Plan end of chemotherapy support	MDT agrees imaging and timing	Agree contacts and surveillance

What each check is for

1 Preparation

Before first infusion

Confirm the exact indication and backbone

Ask / **act:** Complete pathology, molecular, organ-function and consent review.

2 Each cycle

Before each planned dose

Treatment safety can change

Ask / **act:** Review toxicity, results and the active support plan.

3

Surgery assessment

After at most three cycles

Coordinate recovery and local treatment

Ask / **act:** Agree surgery timing; no automatic postoperative nivolumab.

4

Follow-up

During and after surgery

Immune effects may be delayed

Ask / **act:** Keep the immune alert card and report new symptoms promptly.

At the next review, ask

- Which treatment phase am I in, and what happens next?
- Which test is expected next, who arranges it, and who explains the result?
- Who should I contact if an expected appointment has not arrived?

This map does not show whether a test has been ordered, booked or completed. Do not wait for a scheduled test if you develop new symptoms.

Practical treatment planning

Agree the arrangements with your oncology team.

STEP 1

Before the visit

Confirm blood tests, transport, medicine instructions and the oncology contact number.

STEP 2

Time in the unit

Nivolumab is infused over 30 minutes before chemotherapy. Pemetrexed is infused over 10 minutes. Carboplatin is infused over 30 minutes. Blood-result review, preparation, support medicines and observation add to the visit.

STEP 3

Access

Treatment uses suitable intravenous (IV) access. No routine take-home infusion pump is used.

STEP 4

Home support

Take the prescribed folic acid, vitamin B12 and rash-prevention steroid plan; continue folic acid for 21 days after the final pemetrexed dose. Carboplatin dosing uses a verified kidney-function measurement; report dehydration or a change in health.

STEP 5

Surgical review

Keep the oncology and surgical teams informed about new symptoms and other appointments.

Treatment visits

Day-by-day checks

Blood count, kidneys, liver and glucose before each cycle; thyroid tests every 3–6 weeks; cortisol if indicated. Assess symptoms before every dose.

CHAPTER 03

Symptoms and supportive care

IN THIS CHAPTER

Recognising changes, knowing when to call, and using your prescribed supportive-care plan.

Your 21-day cycle.

Timing is a general guide; symptoms may start or stop at other times.

Treatment or check	Timing in this view
All three medicines	 **Day 1** After clinical review
Immune symptoms can occur any day	 **Days 1–21** Nivolumab can inflame healthy organs during treatment or after it stops. Report new symptoms promptly and carry your immune alert card for at least five months after the last dose.

Relative cycle days, not guaranteed symptom or recovery dates.

The next cycle starts 21 days after this one, if your team confirms it.

> Contact your oncology team urgently for fever, feeling suddenly unwell, new breathlessness, chest pain, persistent diarrhoea, severe weakness, confusion, reduced urine or a severe rash. Follow your local emergency instructions; do not wait for the next appointment.

Between-treatment notes

Day 1: Day 1 treatment

All three medicines after review. Confirm symptoms, blood results and support plan.

Day 2–21: At home and review

Effects vary; no guaranteed recovery day. Report new symptoms promptly and prepare for the next assessment.

Support around the medicines

1 Immune alert card

Nivolumab can inflame healthy organs during treatment or after it stops. Report new symptoms promptly and carry your immune alert card for at least five months after the last dose.

2 Prepare for surgery

The oncology and surgical teams agree scans, recovery checks and timing. Report any illness or new symptom before the planned operation.

Recognise effects early

During an infusion

Tell the nurse immediately about chills, rash, flushing, chest discomfort, breathing difficulty or feeling faint.

Everyday care

Plan support around how you feel.

Eating and drinking

Ask for help with nausea, appetite, mouth pain or poor intake. Follow any individual fluid restriction or hydration plan.

Medicines and pregnancy

Tell your team about all medicines and supplements. Discuss fertility and contraception before treatment; pregnancy and breastfeeding require specific advice.

Moving towards surgery

The surgical team decides readiness.

Reassessment

Scans, clinical recovery, organ function and fitness inform the operation plan. Treatment response does not remove the need for surgical review.

Your contacts and questions

Keep your records

Bring the treatment passport, medicines list and immune alert card to appointments or emergency care.

Understanding the plan

Ask your team to confirm the selected backbone, next visit, support medicines and whom to call.

Questions worth asking

Will I receive all five combinations?

No. This guide is for pemetrexed and carboplatin with nivolumab. The five backbones are alternatives; your team prescribes one.

Does nivolumab continue after surgery?

Not automatically under this NCCP regimen. It covers at most three cycles before surgery; any later treatment needs a separate plan.

What if a dose is delayed?

The team reviews blood tests, symptoms and recovery. Do not assume a delay means treatment has failed or that surgery must occur on a fixed date.

> **Try explaining the plan back:** Which medicines are you having now? What changes after induction? What would make you call today? Ask your team to fill any gaps.

Nivolumab

- Immune inflammation in multiple organs
- Infusion reactions
- Delayed effects after treatment stops

Pemetrexed

- Low blood counts, infection and bleeding
- Nausea, vomiting and fatigue
- Mouth soreness and diarrhoea
- Kidney problems
- Allergic or infusion reactions
- Skin rash; rarely serious skin reactions
- Rare serious lung inflammation

Carboplatin

- Low blood counts and infection
- Nausea and vomiting
- Kidney, hearing and nerve effects
- Allergic reactions

Supportive prescription draft for review. This is a regimen template. It is not an issued patient prescription; individual review and completion are required.

Day 1 · pemetrexed / carboplatin with nivolumab

Active medicines: nivolumab + pemetrexed + carboplatin

Emetogenic risk: high

Medicine	Dose / route	When
Aprepitant	125 mg orally Day 1; 80 mg orally Days 2–3	Before systemic anticancer therapy (SACT) on Day 1, then daily on Days 2–3
Ondansetron	16 mg orally	Day 1 Adjust for QT/individual risk.
Antiemetic dexamethasone reference	12 mg orally Day 1; 8 mg orally daily Days 2–4	Dedicated carboplatin AUC ≥4 NCIS V8 row Reference course only: reconcile with pemetrexed rash prophylaxis and aprepitant interaction into one unambiguous prescription. Do not add both courses.
Metoclopramide	10 mg orally three times daily as prescribed	Subsequent days NCIS row does not specify an end day or PRN qualifier; prescriber/pharmacy must specify duration and suitability.
Folic acid	350–1000 micrograms orally daily	At least five doses in seven days before first pemetrexed; throughout treatment and 21 days after the final dose
Vitamin B12 (hydroxocobalamin)	1000 micrograms intramuscular (IM)	Week before first pemetrexed and every three cycles thereafter Later doses may be on the pemetrexed day.
Pemetrexed rash prophylaxis: dexamethasone equivalent	4 mg orally twice daily	Day before, day of and day after pemetrexed This is the regimen rash schedule; prescriber/pharmacy must reconcile it with antiemetic dexamethasone into one unambiguous prescription, including aprepitant interaction. Do not simply add both courses.

Additional phase-specific support

- Take the prescribed folic acid, vitamin B12 and rash-prevention steroid plan; continue folic acid for 21 days after the final pemetrexed dose. Carboplatin dosing uses a verified kidney-function measurement; report dehydration or a change in health.
- Nivolumab can inflame healthy organs during treatment or after it stops. Report new symptoms promptly and carry your immune alert card for at least five months after the last dose.
- The prescriber specifies doses, duration, rescue treatment and interactions; follow one reconciled prescription.

Before signing: verify allergies, current medicines, interactions, renal / hepatic function, QT risk, sedation, prior response, total steroid exposure and the rescue-medicine duration. Do not duplicate medicines from another phase.

Individual changes / rescue duration: _____
Prescriber / date: _____ Pharmacy review: _____

Nivolumab, pemetrexed and carboplatin before surgery

Up to three 21-day cycles. Day 1: nivolumab, pemetrexed and carboplatin. Then surgical multidisciplinary team (MDT) reassessment. The five listed backbones are alternatives; this guide covers only your selected combination.

Source reconciled; independent clinical, pharmacy and patient/PPI review pending. Selected effects; not a complete medicine leaflet.

A picture of side effects

Numbered symptom boxes connect to the anatomical illustration. The medicine responsible may be uncertain; report the symptom.

1 Hormones / energy

Nivolumab

Report severe tiredness, headache, dizziness or unusual thirst.

2 Breathing

Nivolumab

Report new cough, breathlessness or chest pain urgently.

3 Liver

Nivolumab / chemotherapy

Blood tests check liver function; report yellow eyes or dark urine.

4 Bowels

Nivolumab / chemotherapy

Report persistent diarrhoea, blood, pain or inability to drink.

5 Kidneys / urine

Nivolumab / carboplatin

Blood tests check kidney function; report reduced urine.

6 Blood / infection

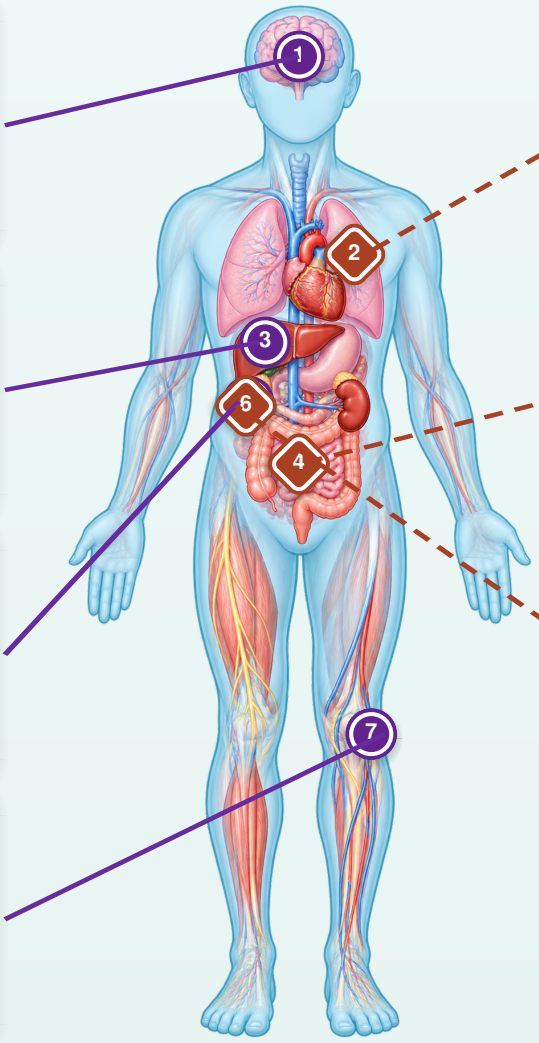
pemetrexed / carboplatin

Fever or sudden illness can be an emergency; follow your contact plan.

7 Skin / hair

pemetrexed / nivolumab

Report a severe rash or blistering urgently; hair changes vary.



Selected effects, not every possible effect or a prediction. Blood-count, skin and line markers are schematic. Number and text carry meaning; colour is decorative.

Contact your oncology team urgently for fever, feeling suddenly unwell, new breathlessness, chest pain, persistent diarrhoea, severe weakness, confusion, reduced urine or a severe rash. Follow your local emergency instructions; do not wait for the next appointment.

CHAPTER 04

Sources and further information

IN THIS CHAPTER

The references behind this information and addresses you can use to find further help.

Use this information alongside the plan agreed with your treatment team. Your team confirms your medicines, doses and appointments.

1. **NCCP 00849 v2 · pNCCP 00849 pp1–7; NCCP 00318 supportive care; NCIS V8 Table 1**
   https://healthservice.hse.ie/documents/6865/849_V2_Nivolumab_chemotherapy.pdf
2. **NCCP 00318 v6 · selected chemotherapy**
   https://healthservice.hse.ie/documents/6921/318_V6_PEMEtrexed_and_CARBOplatin.pdf
3. **EMA Opdivo product information**
   https://www.ema.europa.eu/en/documents/product-information/opdivo-epar-product-information_en.pdf
4. **CheckMate 816 final overall survival**
   https://www.nejm.org/doi/10.1056/NEJMoa2502931
5. **NCCP NCIS antiemetics V8**
   https://healthservice.hse.ie/documents/7180/Supportive_Care_Antiemetics_for_inclusion_NCIS_Medical_Oncology.pdf